AI-Accelerated Discovery of Materials for Carbon Capture and Utilization (CCU)

Author Names: Saksham Walia, Sachin Saji
Affiliations: University of Wollongong India
Email Address: ssw681@uowmail.edu.au, sst818@uowmail.edu.au

Abstract:
This study leverages artificial intelligence to accelerate the discovery of novel nanoporous frameworks for efficient CO_2 capture and catalytic conversion to value-added chemicals, merging quantum simulation data with machine learning to address sustainability in hard-to-abate sectors and advance carbon utilization technologies.

Introduction:

Carbon capture and utilization (CCU) is increasingly recognized as a pivotal global strategy for mitigating anthropogenic CO_2 emissions and achieving deep decarbonization across a diverse range of industrial sectors, including energy, manufacturing, chemicals, and transportation. These sectors, often classified as hard-to-abate, present unique challenges for emission reduction due to legacy infrastructure, complex supply chains, and stringent operational requirements.

The urgent need for effective CCU technologies is underscored by international climate agreements such as the Paris Accord and the United Nations Sustainable Development Goals (SDGs), which emphasize reducing greenhouse gases on a planetary scale to limit global warming and prevent severe climate disruptions. A key enabler in this effort is the identification of advanced materials—including metal-organic frameworks (MOFs), covalent-organic frameworks (COFs), and engineered porous polymers—which exhibit high potential for selective CO_2 capture and efficient catalytic conversion into value-added chemicals like methanol and urea. However, the global scientific community faces persistent obstacles: the immense chemical design space of candidate materials, intricate structure–property relationships, and the slow pace of experimental validation.

Recent breakthroughs in artificial intelligence (AI) and computational materials science are profoundly accelerating progress in this area. AI-driven generative models and high-throughput quantum mechanical simulations, integrated with global experimental databases, empower researchers to predict CO_2 uptake capacities, optimize catalytic activity, and design materials tailored for diverse geographies, climate regimes, and industrial ecosystems. These capabilities not only facilitate the rapid discovery and deployment of multifunctional CCU materials, but also foster cross-border collaboration and knowledge sharing.

This work proposes a comprehensive AI-powered platform aimed at discovering and optimizing high-performance materials for carbon capture and utilization worldwide. By bridging international datasets, simulation tools, and experimental expertise, the platform advances sustainable innovation in direct alignment with SDGs 7 (Affordable and Clean Energy), 9 (Industry, Innovation and Infrastructure), and 13 (Climate Action). The anticipated outcome is a transformative leap towards global decarbonization, supporting nations and industries in achieving net-zero targets and enhancing climate resilience for current and future generations.

Methods:
A data-driven pipeline was constructed, integrating >10,000 quantum simulation records on porous materials, chemisorption characteristics, and catalyst surfaces. Machine learning algorithms—including random forest and deep neural networks—were trained to predict CO_2 uptake and conversion efficiency. Generative adversarial networks (GANs) proposed new framework designs, which were validated using density functional theory (DFT) simulations, and a subset was synthesized and experimentally tested for CO_2 conversion rates

Results and Discussion:
The pipeline identified multiple novel MOF structures predicted to surpass benchmark materials in CO_2 adsorption and catalytic conversion. For example, GAN-designed MOF-1263 exhibited a simulated uptake of 7.9 mmol/g—30% higher than current references—and DFT showed favorable active sites for CO_2 hydrogenation. Initial synthesis trials confirmed stability and reproducible performance under simulated flue gas conditions. Comparative life cycle assessments indicated a potential reduction in net CO_2 emissions of 47% for cement and ammonia plants incorporating these materials. Discussion covers how the AI model improved design efficiency, reduced false positives, and accelerated lab validation. Challenges include model transferability to real-world environments, scalability of frameworks, and integration with process engineering under techno-economic constraints. Multi-objective optimization illustrated the trade-offs between capture capacity, conversion yield, material cost, and environmental resilience.

Significance and Impact:
By fusing AI with materials discovery and CCU, this work delivers efficient pathways for industrial decarbonization. The approach enables rapid innovation and sustainable solutions for hard-to-abate sectors, supporting climate resilience and global emissions reduction.

References:

Li, X., et al. "Machine Learning in MOF Discovery for Carbon Capture." Nat. Mater., 2023.

UNFCCC. *Paris Agreement*. United Nations Framework Convention on Climate Change, 2015.

United Nations. *Sustainable Development Goals Report 2025*. UN, 2025.

International Energy Agency (IEA). *CCUS in Clean Energy Transitions*. IEA Report, 2022.

Zhou, C., et al. *Artificial intelligence for materials discovery in the chemical sciences*. **Chem. Soc. Rev.**, 50, 3490–3520 (2021).